6. Show that the matrices $A=\begin{pmatrix}1 & 2\\ 3 & 1\end{pmatrix}$, $B=\begin{pmatrix}1 & -2\\ -3 & 1\end{pmatrix}$ satisfy commutative property $AB=BA$

7. Let $A=\begin{pmatrix}1 & 2\\ 1 & 3\end{pmatrix}$, $B=\begin{pmatrix}4 & 0\\ 1 & 5\end{pmatrix}$, $C=\begin{pmatrix}2 & 0\\ 1 & 2\end{pmatrix}$ Show that (i) $A(BC)=(AB)C$
(ii) $(A-B)C=AC-BC$ (iii) $(A-B)^T=A^T-B^T$

8. If $A=\begin{pmatrix}\cos\theta & 0\\ 0 & \cos\theta\end{pmatrix}$, $B=\begin{pmatrix}\sin\theta & 0\\ 0 & \sin\theta\end{pmatrix}$ then show that $A^2+B^2=I$.

9. If $A=\begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}$ prove that $AA^T=I$.

10. Verify that $A^2=I$ when $A=\begin{pmatrix}5 & -4\\ 6 & -5\end{pmatrix}$

11. If $A=\begin{pmatrix}a & b\\ c & d\end{pmatrix}$ and $I=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$ show that $A^2-(a+d)A=(bc-ad)I_2$

12. If $A=\begin{pmatrix}5 & 2 & 9\\ 1 & 2 & 8\end{pmatrix}$, $B=\begin{pmatrix}1 & 7\\ 1 & 2\\ 5 & -1\end{pmatrix}$ verify that $(AB)^T=B^TA^T$

13. If $A=\begin{pmatrix}3 & 1\\ -1 & 2\end{pmatrix}$ show that $A^2-5A+7I_2=0$

Exercise 3.20

6ZM3R

Multiple choice questions

1. A system of three linear equations in three variables is inconsistent if their planes
(A) intersect only at a point (B) intersect in a line
(C) coincides with each other (D) do not intersect

2. The solution of the system $x+y-3z=-6$, $-7y+7z=7$, $3z=9$ is
(A) $x=1, y=2, z=3$ (B) $x=-1, y=2, z=3$
(C) $x=-1, y=-2, z=3$ (D) $x=1, y=-2, z=3$

3. If $(x-6)$ is the HCF of $x^2-2x-24$ and x^2-kx-6 then the value of k is
(A) 3 (B) 5 (C) 6 (D) 8

4. $\dfrac{3y-3}{y}\div\dfrac{7y-7}{3y^2}$ is
(A) $\dfrac{9y}{7}$ (B) $\dfrac{9y^3}{(21y-21)}$ (C) $\dfrac{21y^2-42y+21}{3y^3}$ (D) $\dfrac{7(y^2-2y+1)}{y^2}$

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